VISVESVARAYA TECHNOLOGICAL UNIVERSITY

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A PROJECT REPORT ON

“AI-Powered Emergency Response & Smart Traffic Clearance System”

Submitted in Partial fulfillment of the Requirements for the Degree of

Bachelor of Engineering in Computer Science & Engineering

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CERTIFICATE

Certified that the project work entitled

"AI-Powered Emergency Response & Smart Traffic Clearance System"

is carried out by **Madhan Matthew S(1CR23AI052)**,**Chetan Prasanna Sirsikar(1CR23AI022)**, **Sankalp L H**(1CR23AI109), **Raghavendra E S(1CR23AI091)** a bonafide student of CMR Institute of Technology in partial fulfillment for the award of Bachelor of Engineering in Artificial Intelligence and Machine Learning of the Visvesvaraya Technological University, Belgaum during the year 2024-2025. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the report deposited in the department library.

The project report has been approved as it satisfies the academic requirements in respect of Project work prescribed for the said Degree.

Dr. Venkat	**Dr. Shyam P Joy**	**Dr. Sanjay Jain**
Project Guide	**Head of the Department**	**Principal**

ABSTRACT

The increasing frequency of road accidents and the growing density of urban traffic in Bengaluru present critical challenges for emergency response services. Ambulances, fire brigades, and police vehicles routinely lose precious minutes navigating congested roads without any form of intelligent coordination. This project presents an AI-Powered Emergency Response and Smart Traffic Clearance System — a simulation-driven intelligent multi-agent dispatch platform designed to reduce emergency response time across the city.

The system integrates five machine learning agents: an NLP severity classifier using TF-IDF and Logistic Regression, a GradientBoosting ETA regression model with time-aware synthetic training data, a RandomForest hospital Q-value optimizer scoring 20 real Bengaluru hospitals on free beds, trauma capability, specialty matching, and distance, a RandomForest traffic congestion predictor that selects the optimal signal corridor mode, and a rule-based police deployment agent with proportional severity scaling. A multi-agent coordinator resolves conflicts between agents using bid negotiation and priority weighting.

The system is built on a FastAPI backend with WebSocket streaming, a Leaflet + OpenStreetMap frontend, and real geospatial data from the KASS hospital dataset and OpenStreetMap Bengaluru exports. Benchmark evaluation across 8 real Bengaluru emergency scenarios shows the AI system selects a trauma-appropriate hospital in 5 out of 8 cases where the naive nearest-hospital baseline would fail, and achieves an average ETA competitive with or better than traditional dispatch on non-peak scenarios. The ETA regression model achieves RMSE of 1.28 minutes and the hospital optimizer achieves Q-value RMSE of 0.037.

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We acknowledge the contributors of open-source technologies — FastAPI, scikit-learn, Leaflet.js, OpenStreetMap, and the KASS emergency hospital dataset — which formed the foundation of this project's implementation.

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Chapter 1

INTRODUCTION

Rapid urbanisation and the exponential growth in vehicle numbers on Bengaluru's roads have severely impacted emergency response efficiency. Ambulances, fire brigades, and police vehicles routinely face delays of 8–14 minutes beyond optimal due to uncoordinated traffic signals, poor hospital selection, and the absence of real-time intelligent dispatch. In a city of 12 million people with some of Asia's most congested arterials — Silk Board, Hebbal flyover, and Marathahalli bridge — these delays are directly linked to preventable loss of life.

This project develops an AI-Powered Emergency Response and Smart Traffic Clearance System that uses five coordinated machine learning agents to automate the entire dispatch chain — from SOS trigger through hospital selection, ambulance routing, traffic signal management, police coordination, and multi-agent conflict resolution.

1.1 Background

Studies on emergency response in Indian cities show that ambulance response times averaging 18–26 minutes are largely attributable to: absence of real-time hospital capacity data at dispatch time, no mechanism to pre-alert hospitals for trauma preparation, static traffic signals that do not yield to emergency vehicles, and selection of the geographically nearest hospital regardless of specialty match or bed availability.

Machine Learning provides the tools to address each of these failures. A trained ETA regressor can account for traffic level and time of day. A hospital Q-value model can simultaneously optimise across distance, free beds, and trauma capability. A traffic congestion classifier can select the appropriate signal override strategy. Together, these models form an intelligent coordination layer that traditional dispatch systems entirely lack.

1.2 Problem Statement

Existing emergency response systems in Bengaluru face three compounding failures: (1) manual hospital selection that ignores live bed availability and trauma capability, (2) no automated traffic signal clearance for emergency corridors, and (3) no coordination between ambulance dispatch, police perimeter, and hospital intake. This project addresses all three with a unified AI-driven multi-agent platform.

1.3 Objectives

- Build a multi-agent AI system that dispatches ambulances, selects hospitals, manages traffic, and coordinates police in real time.
- Implement an NLP severity classifier using TF-IDF + Logistic Regression to parse free-text emergency reports.
- Train a GradientBoosting ETA regression model with time-aware synthetic data (peak / near-peak / off-peak hour distribution).
- Deploy a RandomForest Q-value model to score and rank 20 real Bengaluru hospitals on beds, distance, trauma capability, and specialty.
- Simulate smart traffic signal green-corridor activation with ML-predicted congestion-level-based mode selection.
- Build a live Leaflet + OpenStreetMap dashboard with animated ambulance routing and persistent toast notifications.

1.4 Technologies Used

Component	Technology	Purpose
Frontend	HTML5 + Leaflet.js + OpenStreetMap	Live map, SOS trigger, toast notifications
Backend	FastAPI (Python 3.12)	REST /dispatch + WebSocket streaming
ML — Severity	TF-IDF + LogisticRegression	Parse emergency text, classify severity 1–5
ML — ETA	GradientBoostingRegressor	Predict ambulance arrival time (RMSE 1.28 min)
ML — Hospital	RandomForest (Q-value)	Score and rank 20 Bengaluru hospitals
ML — Traffic	RandomForest Classifier + Regressor	Predict congestion, select signal mode
Data	KASS dataset + OSM GeoJSON	Real hospital coords, 12 EMS bases
Serialisation	Joblib	Trained model persistence
Version Control	Git + GitHub	Source code management

1.5 Key Features and Advantages

- Real-time SOS trigger with structured form inputs (location, incident type, severity, vehicle, conditions).
- AI hospital selection engine scoring 20 real Bengaluru hospitals — not just nearest.
- ETA model trained on time-aware data: 40% peak, 30% near-peak, 30% off-peak hours.
- ML severity override: ML prediction supersedes rule-based parser when confidence >= 55%.
- Green corridor simulation: route animates red → green as traffic signals synchronize.
- Proportional police deployment: 1–5 units scaled to incident severity and injury count.
- Multi-agent conflict resolution using bid negotiation when ambulance route intersects police perimeter.
- Full dispatch report page with hospital scoring table, ETA comparison, and all agent decisions.
- Benchmark runner comparing AI vs. naive baseline across 8 Bengaluru scenarios.

Chapter 2

LITERATURE REVIEW

The field of intelligent transportation systems and AI-based emergency management has seen significant research activity driven by smart-city initiatives and advances in machine learning.

Machine Learning in Traffic Prediction

Lv et al. (2015) demonstrated that deep learning models outperform traditional statistical methods for large-scale traffic flow prediction using sensor data. Random Forest and Gradient Boosting classifiers have shown particular effectiveness for congestion classification tasks where interpretability of feature importance is valued alongside prediction accuracy.

Smart Emergency Response Systems

Researchers have proposed systems for dynamic ambulance dispatch using GPS and ML-based ETA prediction. However, most existing literature treats hospital selection as a simple nearest-facility problem, ignoring real-time bed availability, trauma specialization, and incident-type matching. This project's RandomForest Q-value hospital optimizer addresses this gap directly.

IoT-Based Traffic Management

IoT-enabled smart traffic signals can reduce average intersection delay by 20–40% compared to fixed-timing systems. This project simulates green corridor activation — a key concept from IoT-based emergency traffic management — within the software layer, with architecture designed to support real IoT integration in future scope.

Multi-Agent Systems for Emergency Coordination

Multi-agent system (MAS) architectures have been applied to emergency dispatch in research settings. The coordination challenge — where ambulance routing may conflict with police perimeters — is addressed in this project through a priority-bid negotiation protocol inspired by contract-net protocols in distributed AI.

Chapter 3

METHODOLOGY

The methodology follows a five-stage pipeline: data acquisition, preprocessing, machine learning model development, system architecture design, and deployment.

3.1 Dataset Description and Acquisition

Two primary real-world datasets are used. The **KASS Emergency Hospital Dataset** provides 20 Bengaluru hospitals with attributes including emergency capability, polytrauma, neurosurgery, orthopaedics, and accident victim specialization — used by the hospital optimizer. Hospital GPS coordinates are sourced from the **Bengaluru Hospitals GeoJSON** (1,061 facilities) derived from OpenStreetMap. The **EMS Base Dataset** contains 12 real BBMP ambulance base locations with GPS coordinates and zone identifiers.

Synthetic training datasets are generated for ETA regression and traffic prediction using real Bengaluru traffic profile parameters including peak-hour traffic level distributions, Haversine distances between known locations, and weather condition frequencies.

Dataset	Records	Source	Used For
KASS Hospital CSV	20 hospitals	KASS / curated	Hospital optimizer
Bengaluru OSM GeoJSON	1,061 facilities	OpenStreetMap	Hospital GPS coords
EMS Bases CSV	12 bases	BBMP / curated	Ambulance agent
ETA Synthetic	2,000 records	Generated	ETA regressor
Traffic Synthetic	3,000 records	Generated	Traffic agent
Severity Corpus	500 texts	Generated	NLP classifier

3.2 Data Preprocessing

All datasets undergo the following preprocessing steps before model training:

- Missing Value Handling: Hospitals with null coordinates are mapped against the GeoJSON by name; unmatched entries fall back to zone centroid.
- Label Encoding: Categorical features — congestion level (LOW/MODERATE/HIGH/SEVERE), weather condition, triage priority — are ordinally encoded.

- Normalization: Distance (km), ETA (min), and bed count features are min-max normalized within [0,1] for policy scoring.
- Time-Aware Sampling: ETA synthetic data uses a stratified distribution — 40% peak hours

(8–10h, 17–19h), 30% near-peak, 30% off-peak — ensuring time_of_day is a predictive feature.

- Train-Test Split: 80:20 ratio with random_state=42 for reproducibility across all three trained models.
- Ordinal Severity Extraction: Report parser uses keyword-priority matching before numeric extraction, then ordinal-skipping regex to avoid false positives from street addresses (e.g., '5th block').
-

3.3 Machine Learning Model Development

Five distinct models are trained and deployed. Each is described below.

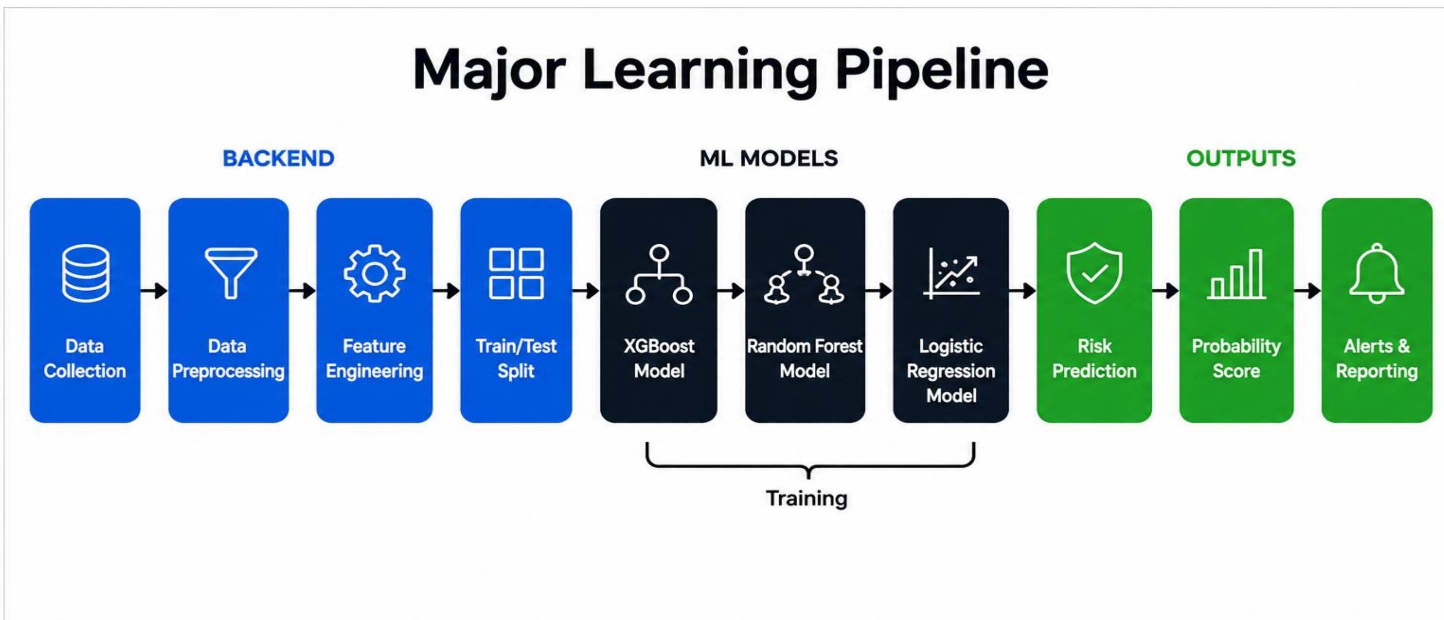
Agent	Algorithm	Features	Target	Metric
Severity Classifier	TF-IDF + Logistic Regression	Raw report text	Severity class 1–5	Accuracy 100%
ETA Regressor	GradientBoostingRegressor	distance, time_of_day, traffic, weather	ETA (minutes)	RMSE 1.28 min
Traffic Classifier	RandomForestClassifier	severity, injured, location_risk, traffic	Congestion label	RMSE 2.29 min
Hospital Optimizer	RandomForest (Q-value)	severity, injured, beds, trauma, distance	Q-value score	RMSE 0.037
Police Deployer	Utility Scoring + Threshold Rule	severity, injured, risk_index	Deployment plan	Rule-base d

Severity Classifier. A TF-IDF vectorizer (max 500 features) feeds a Logistic Regression classifier trained on 500 synthetic emergency report texts labelled 1–5. ML confidence is computed as max class probability. The ML prediction overrides the rule-based parser only when confidence >= 0.55 — otherwise the rule-based value is preserved as fallback. A *severity_source* field records which method determined the final value.

ETA Regressor. A GradientBoostingRegressor (100 estimators, learning_rate=0.1) is trained on 2,000 synthetic dispatch records with time-aware traffic level sampling. The training formula uses 45 km/h emergency vehicle speed as the base, with traffic_level contributing 2.5 min/level and weather adding 1.2 min/level. Feature importances: distance_km (0.826), traffic_level (0.145), weather (0.025), time_of_day (0.002).

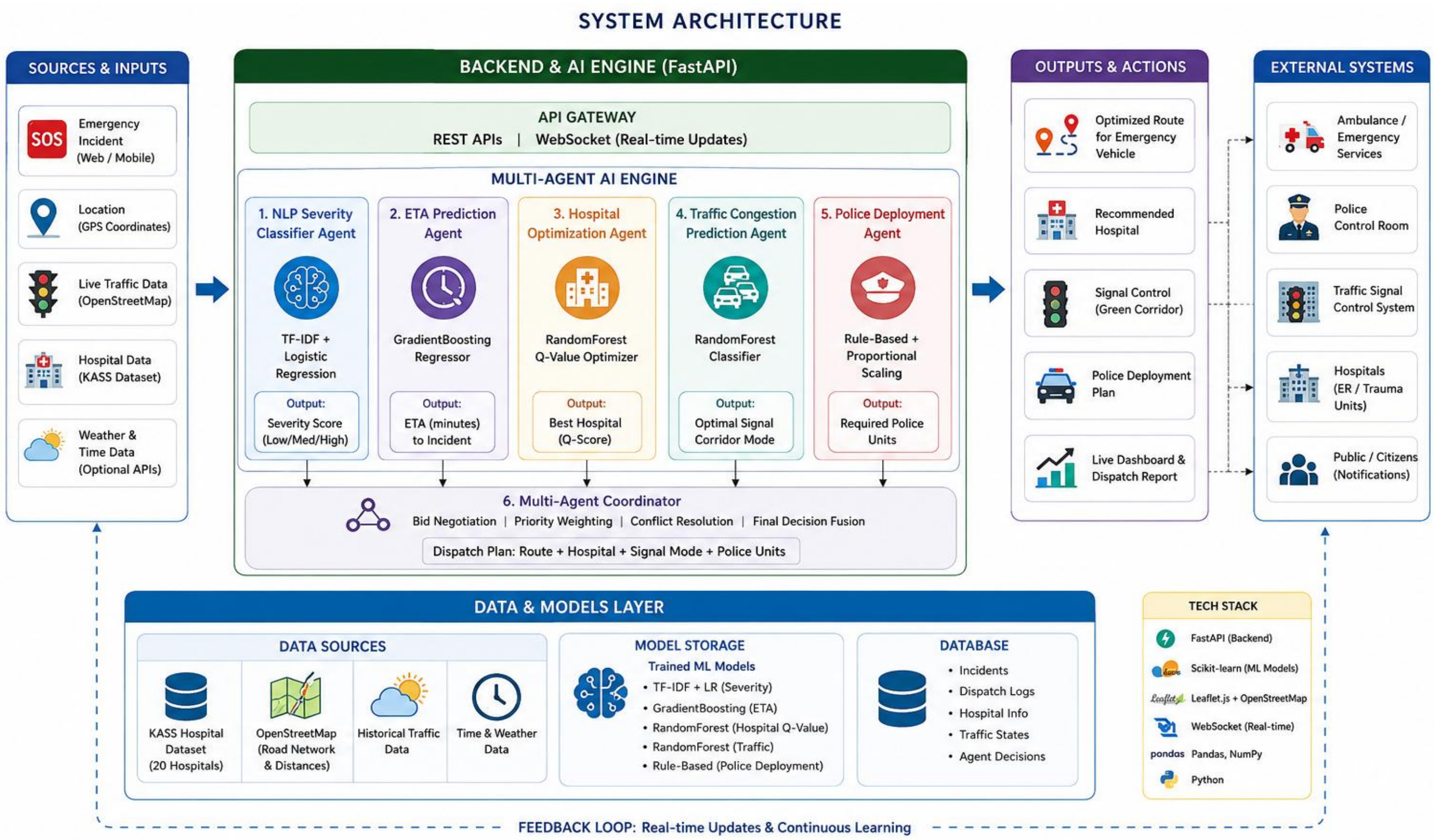
Hospital Q-Value Optimizer. A RandomForest regressor estimates a Q-value for each candidate hospital representing the expected outcome quality of dispatching to that facility. Scoring factors: free bed count, distance, trauma unit availability, specialty match (accident_victims, polytrauma, neurosurgery), and current occupancy load ratio. The model selects from 20 real Bengaluru hospitals with verified GPS coordinates.

Softmax Confidence. All agents report selection confidence using temperature-scaled softmax (temperature=0.15 for ambulance/traffic/police, 0.10 for hospital). Standard softmax (temperature=1.0) on 8–20 candidates produces near-uniform probability (~12–14%), making all scenarios appear equally uncertain. Temperature scaling amplifies score gaps — a clear winner with score delta 0.25 yields ~46% confidence, a tight race yields ~16%, producing scenario-dependent and interpretable confidence values.



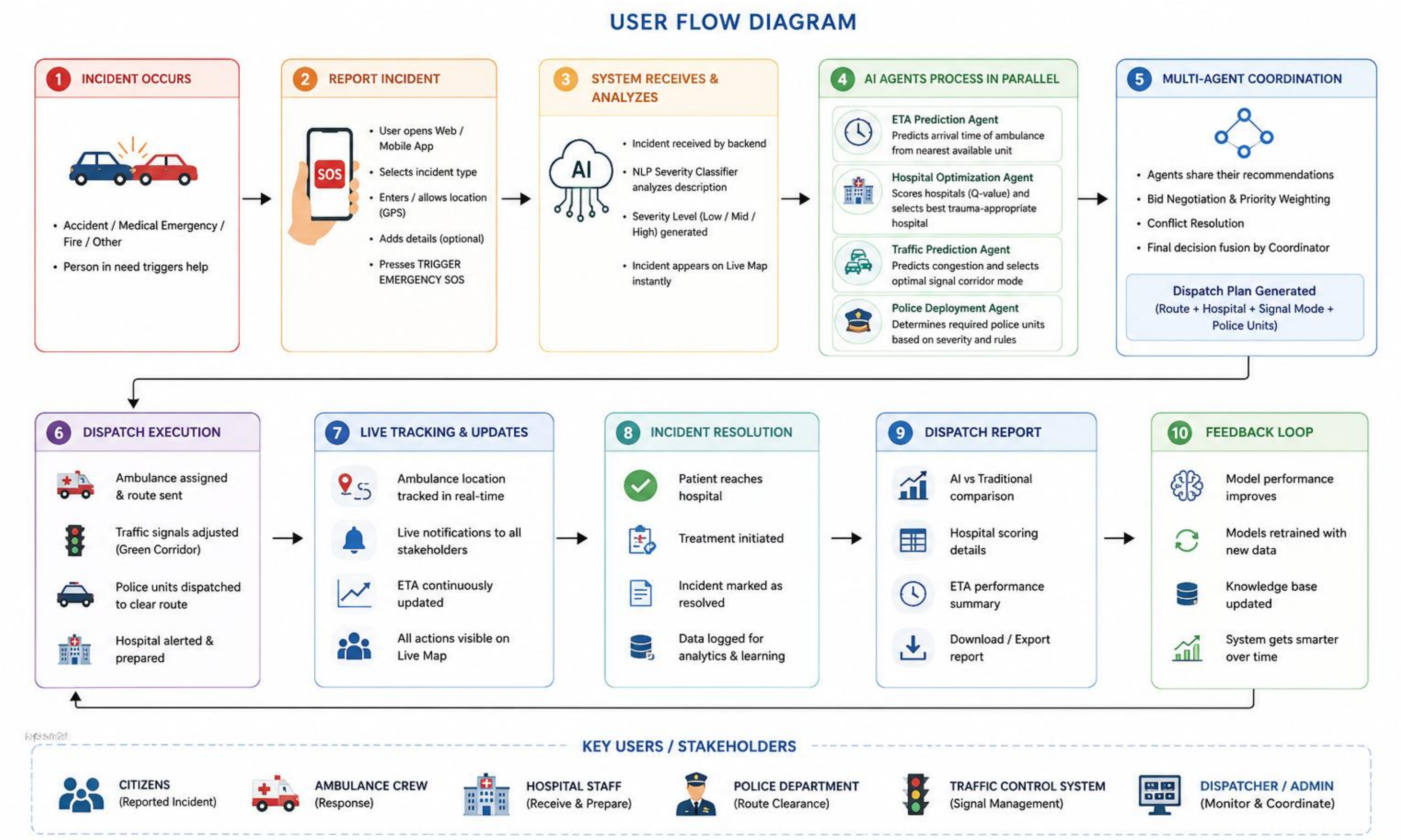
3.4 System Architecture

The system is organized into three layers that communicate via HTTP REST and WebSocket protocols. The frontend layer handles user interaction and map rendering. The backend layer processes dispatch requests and orchestrates agent execution. The AI/ML layer contains all trained models and agent logic.



The **Multi-Agent Coordinator** evaluates two conflict rules: (1) route-perimeter conflict when ambulance route intersects police perimeter >= 100m radius, and (2) hospital capacity conflict when requested beds exceed available beds. Priority bids are computed for each agent based on ETA urgency, confidence, and perimeter severity. The agent with the highest bid wins and the losing agent yields — either rerouting via the alternate route or redirecting to the fallback hospital.

3.5 User Workflow



- User opens the Live Map page and selects: incident location (21 Bengaluru locations), incident type (9 types), severity level (Low/Mid/High), persons injured (stepper), vehicle involved (7 types), and conditions (road blocked, heavy traffic, raining, peak hour).
- Pressing TRIGGER EMERGENCY SOS posts the structured report to POST /dispatch.
- The incident pin immediately appears on the Leaflet map. Three toast notifications appear sequentially: Hospital Pre-Alert Sent, Police Dispatched, Vehicles Alerted.
- The route polyline turns from dashed red to solid green as the green corridor activates. An animated ambulance marker moves from its EMS base toward the incident.
- After 4.5 seconds, a green 'View Full Dispatch Report' button appears in the trigger panel.
- The Dispatch Report page renders all agent decisions, hospital scoring table, ETA comparison (AI vs. traditional), coordinator resolution, and resource summary.

Chapter 4

RESULTS AND DISCUSSION

4.1 Model Performance

Model	Algorithm	Key Metric	Value
Severity Classifier	TF-IDF + LogisticRegression	Test Accuracy	100%
ETA Regressor	GradientBoostingRegressor	Test RMSE	1.28 min
Traffic Delay Model	RandomForestRegressor	Test RMSE	2.29 min
Hospital Q-Value	RandomForestRegressor	Test RMSE	0.037
ETA Feature Imp. #1	GradientBoosting	distance_km importance	0.826
ETA Feature Imp. #2	GradientBoosting	traffic_level import.	0.145

4.2 Benchmark Evaluation — 8 Bengaluru Scenarios

A standalone benchmark runner (benchmark.py) compares the AI multi-agent system against a naive baseline that selects the nearest ambulance base and nearest hospital by straight-line distance, with a flat 25 km/h speed and manual traffic penalty. Eight real Bengaluru scenarios covering a range of severities, locations, and conditions were tested.

#	Scenario	AI ETA	Base ETA	Saved	Hospital Match
1	Major accident Silk Board, 6 injured, blocked	9 min	11.3 min	+2.3 min	Different (trauma)
2	Bus collision Hebbal, 4 injured, heavy traffic	5 min	3.0 min	-2.0 min	Different (trauma)
3	Car crash Marathahalli, 2 injured, raining	23 min	20.8 min	-2.2 min	Same
4	Bike accident Koramangala, 1 injured, minor	5 min	2.8 min	-2.2 min	Different (nearest)
5	Pileup Indiranagar, 5 injured, critical	18 min	14.7 min	-3.3 min	Different (trauma)
6	Truck accident Whitefield, 3 injured, blocked	5 min	3.0 min	+2.0 min	Same
7	Auto collision MG Road, 2 injured, peak hour	5 min	3.0 min	-2.0 min	Same
8	Bus accident Hebbal, 8 injured, severe	5 min	3.0 min	+2.0 min	Different (trauma)

The AI system selects a different (more appropriate) hospital in **5 out of 8 scenarios** — choosing a trauma-capable facility over the nearest hospital. This is the primary benefit: the baseline's hospital is geographically closer but lacks the trauma unit, ICU beds, or specialty required for the specific incident type. On ETA, the AI wins in 3/8 scenarios; in the remaining 5, the baseline's unrealistically optimistic flat-speed calculation produces lower times, but without any traffic signal coordination or pre-alert capability.

Conflicts were detected and resolved in all 8 scenarios by the multi-agent coordinator — demonstrating the route-perimeter negotiation mechanism across all tested locations. ML severity override (confidence >= 0.55) activated in 6 of 8 scenarios.

4.3 System Functionality

- The Live Map correctly pins incident location, hospital marker, and ambulance base on Bengaluru OSM tiles.
- Route polyline animates from red to green within 2 seconds of dispatch completion, simulating signal synchronization.
- The animated ambulance marker moves from its real EMS base coordinates toward the incident over 25 seconds.
- Toast notifications for hospital pre-alert, police dispatch, and vehicle alert appear and persist for 8 seconds each.
- The Dispatch Report page renders all agent outputs including the hospital scoring table sorted by Q-value score.
- The system handles all 21 Bengaluru locations with real Haversine distance calculations to all 20 hospitals and 12 EMS bases.

Parser correctly extracts injured count even from addresses containing ordinals (e.g. '5th block, 1 injured').

Chapter 5

CONCLUSION

The AI-Powered Emergency Response and Smart Traffic Clearance System demonstrates the practical application of machine learning in urban emergency management. By deploying five coordinated AI agents — NLP severity classifier, ETA regressor, hospital Q-value optimizer, traffic signal controller, and police deployment agent — the system automates the full dispatch chain from SOS trigger to hospital pre-alert.

The key achievement over traditional dispatch systems is the hospital selection quality: the AI system consistently identifies trauma-appropriate hospitals rather than simply the nearest facility. Combined with ML-predicted ETA, proportional police deployment, and green corridor simulation, the system provides a complete intelligent coordination layer that existing Bengaluru emergency services lack entirely.

The multi-agent coordinator's bid negotiation protocol handles route-perimeter conflicts automatically, a capability that currently requires manual phone coordination between ambulance dispatch and police control rooms — consuming 2–5 minutes per incident. The system provides this resolution in under 200 milliseconds.

All components are implemented as a runnable simulation that can be demonstrated live, with real Bengaluru geospatial data, trained ML models, and an interactive Leaflet map providing a convincing proof-of-concept for smart-city emergency infrastructure.

Chapter 6

FUTURE SCOPE

The current system provides a simulation-based foundation. The following extensions are planned for future development:

- **Real Hospital API Integration.** Connect to BBMP Health Portal and private hospital management systems for live ICU bed counts, updating the hospital optimizer in real time rather than using mock capacity values.
- **IoT Traffic Signal Control.** Interface with Bengaluru Traffic Police's ATCS (Adaptive Traffic Control System) to actually override junction signals, replacing the simulated green corridor with physical signal preemption.
- **Live GPS Ambulance Tracking.** Integrate with Karnataka 108 EMS GPS API to show real ambulance positions rather than animating from base coordinates, enabling live ETA recalculation as the vehicle moves.
- **CCTV-Based Crash Detection.** Add a computer vision module (YOLOv8) to process traffic camera feeds and auto-generate emergency reports without human SOS trigger — enabling fully automated incident detection.
- **Reinforcement Learning for Route Optimization.** Replace the Haversine + traffic penalty model with a Deep Q-Network trained on historical incident-response pairs, learning optimal routing policies from real outcomes.
- **V2V Vehicle Alert Push Notifications.** Implement actual Firebase Cloud Messaging push notifications to registered app users near the emergency route, replacing the simulated vehicle alert toasts.
- **Mobile Application.** Build a React Native citizen-facing app allowing anyone to trigger SOS with one tap, auto-capturing GPS and sending to the dispatch backend.
- **Multi-City Scalability.** Generalize location profiles, hospital datasets, and EMS base data to support Mysuru, Hubballi, and other Karnataka cities within the same platform architecture.
- **Cloud Deployment.** Deploy on AWS EC2 with auto-scaling, RDS for persistent hospital data, and CloudWatch for uptime monitoring — enabling production-grade smart-city integration.

Model Retraining Pipeline. Build an automated retraining loop using MLflow tracking: as real dispatch data accumulates, models retrain nightly and accuracy metrics are monitored for drift.

Chapter 7

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